

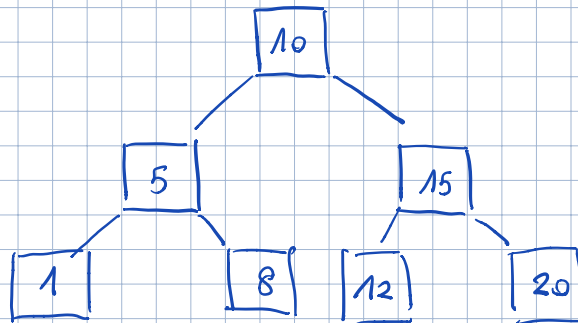
Esempio

$t=2$

1, 2, 3

$h=2$

Numero minimo di chiavi



$n=7$

$x.m = 1$

$\rightsquigarrow$

10

$y.m = 3$

$\rightsquigarrow$

1 | 5 | 8

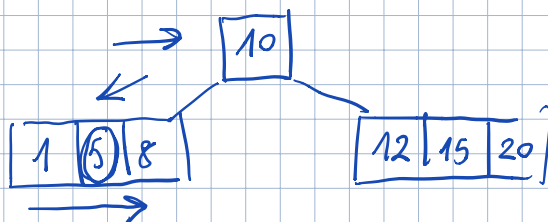
12 | 15 | 20

$n=7$

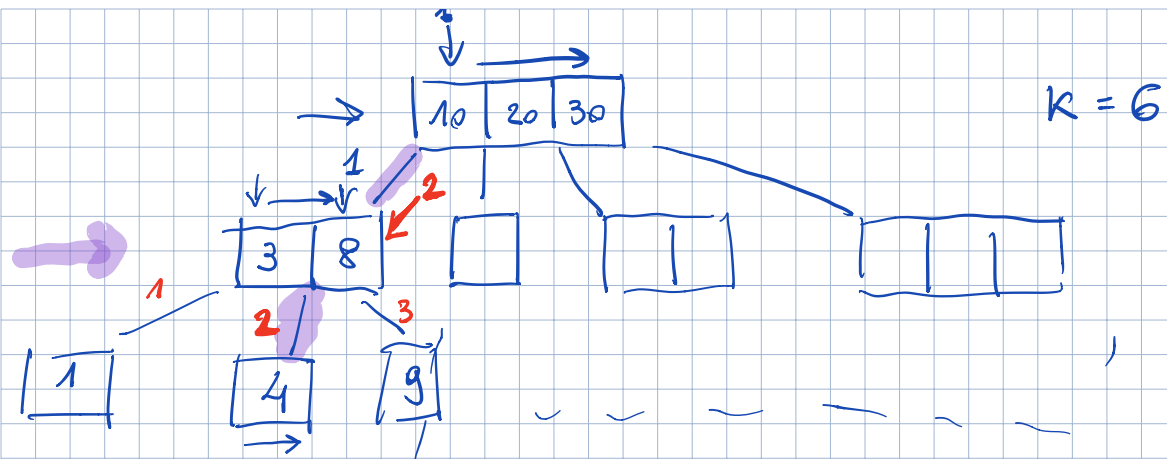
$h=1$

Ricerca

T t k cercare k in T

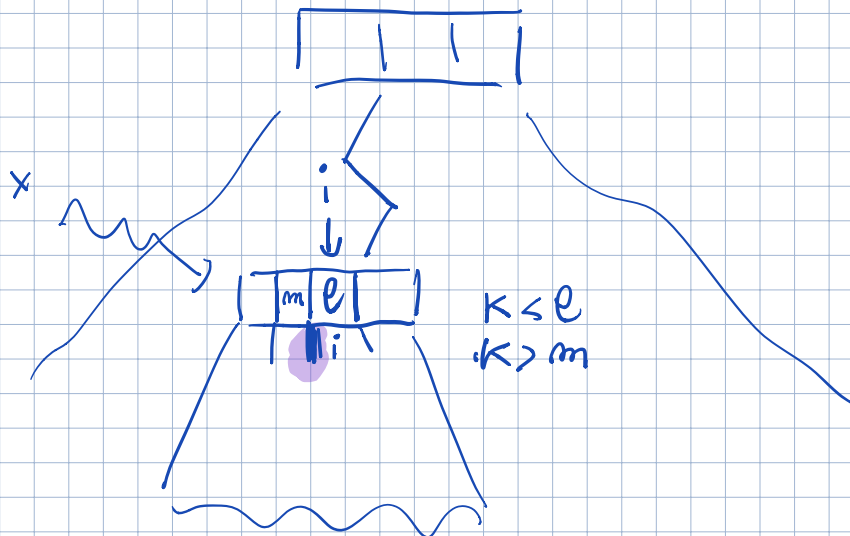


$k=5$



$K=6$ non è in T

Idea



$x.key_{x.m}$

$x.key[x.m]$

BtreeSearch (T, t, x, k) $\quad \nexists \frac{x}{e} \text{ in RAM}$

```

    i ← 1
    while ( i ≤ x.m && x.keyi < k ) {
        i ← i + 1
    }
    if ( i ≤ x.m && x.keyi = k ) {
        return ( x, i )
    } else {
        if ( x.leaf ) {
            return NIL
        } else {
            ⇒ y ← DiskRead ( x.ci )
            return BtreeSearch ( T, t, y, k )
        }
    }
}

```

y existe sempre? $i = x.m + 1$ si

CPU

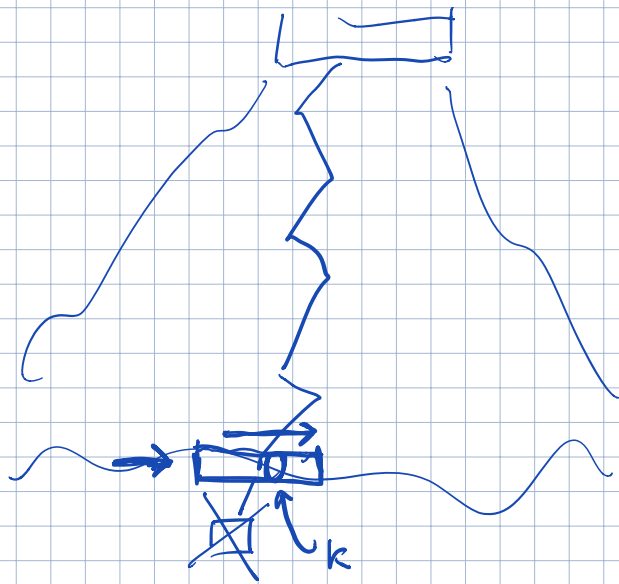
$$O(h \cdot t) = O(t \cdot \log_t m)$$

R/W

$$O(h) = O(\log_t m)$$

Nota: Non ho il puntatore $x.parent$
da x posso solo scendere

Inserimento



K

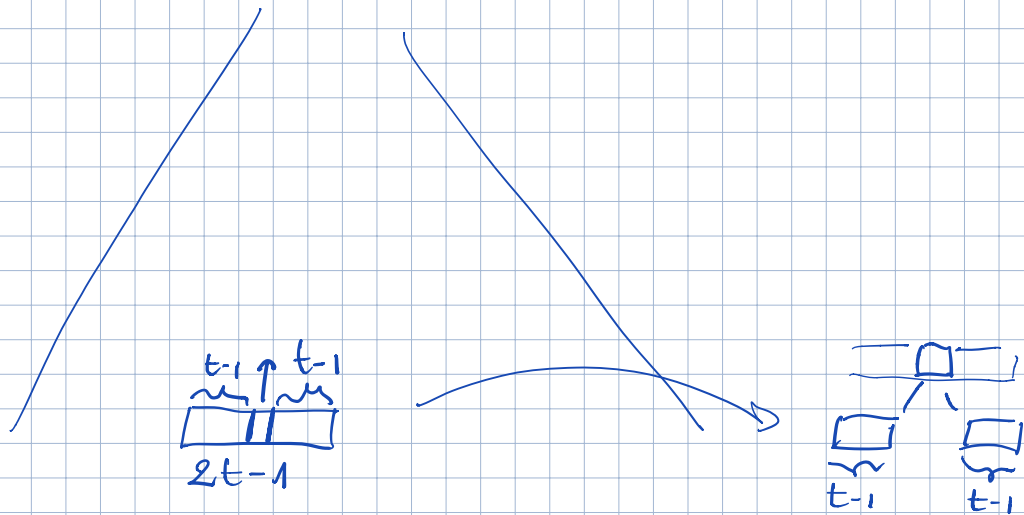
K viene inserito
in una foglia
che esiste
già

Problema

Cosa faccio se la foglia è già piena?

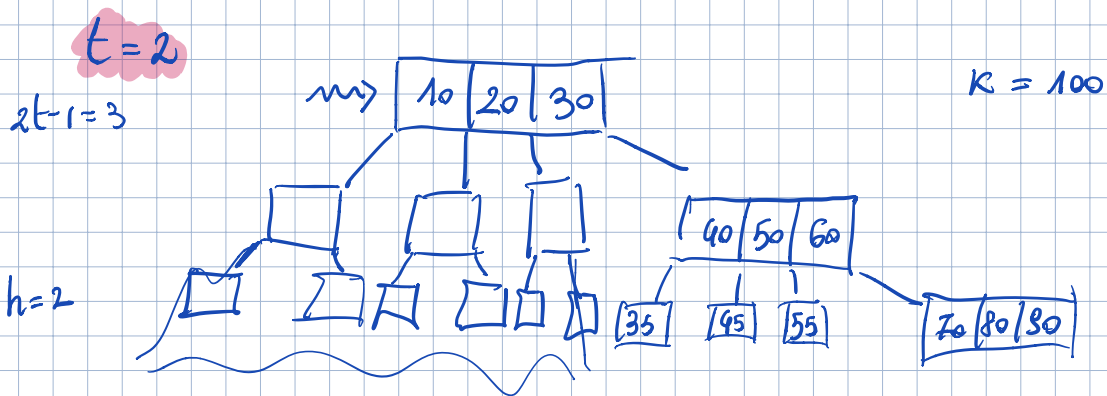
$$x.n = 2t - 1 \quad ?$$

Split $\leadsto$ da un nodo ne
genero due

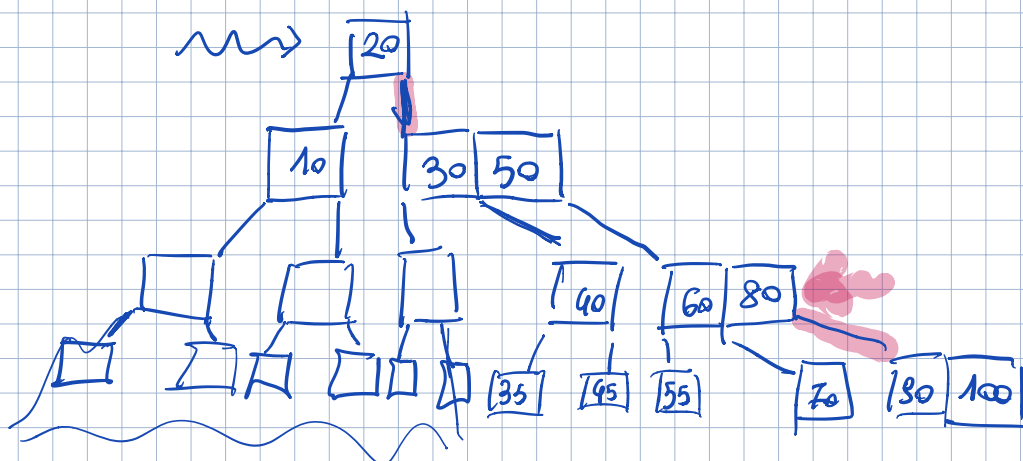
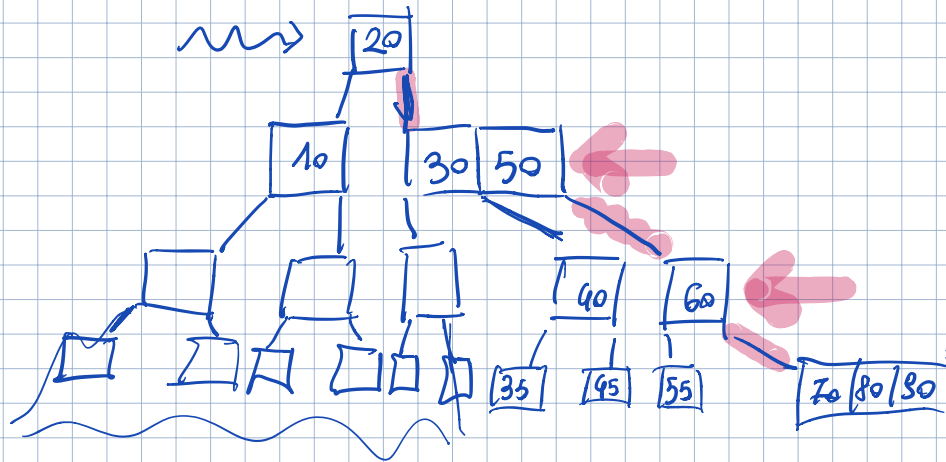
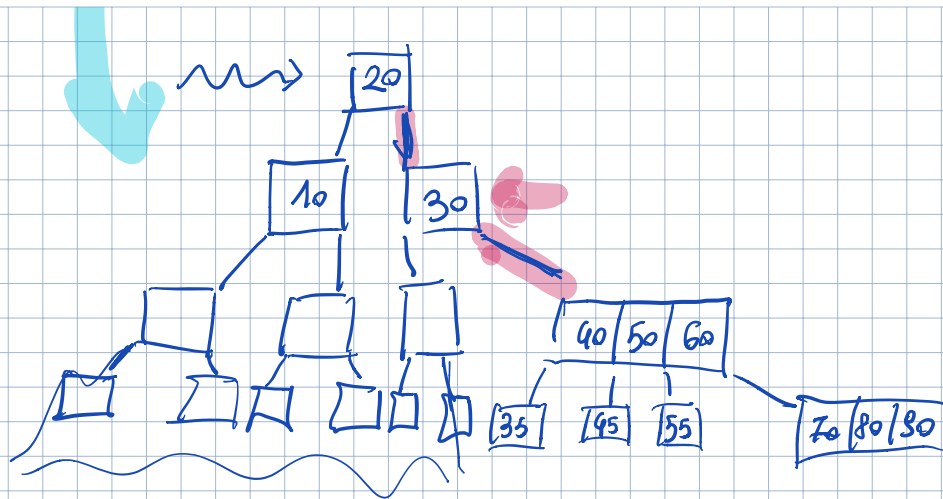


Eseguiremo gli split secondo
dalla radice verso le foglie
ogni volta che incontreremo un nodo
pieno

Esempio



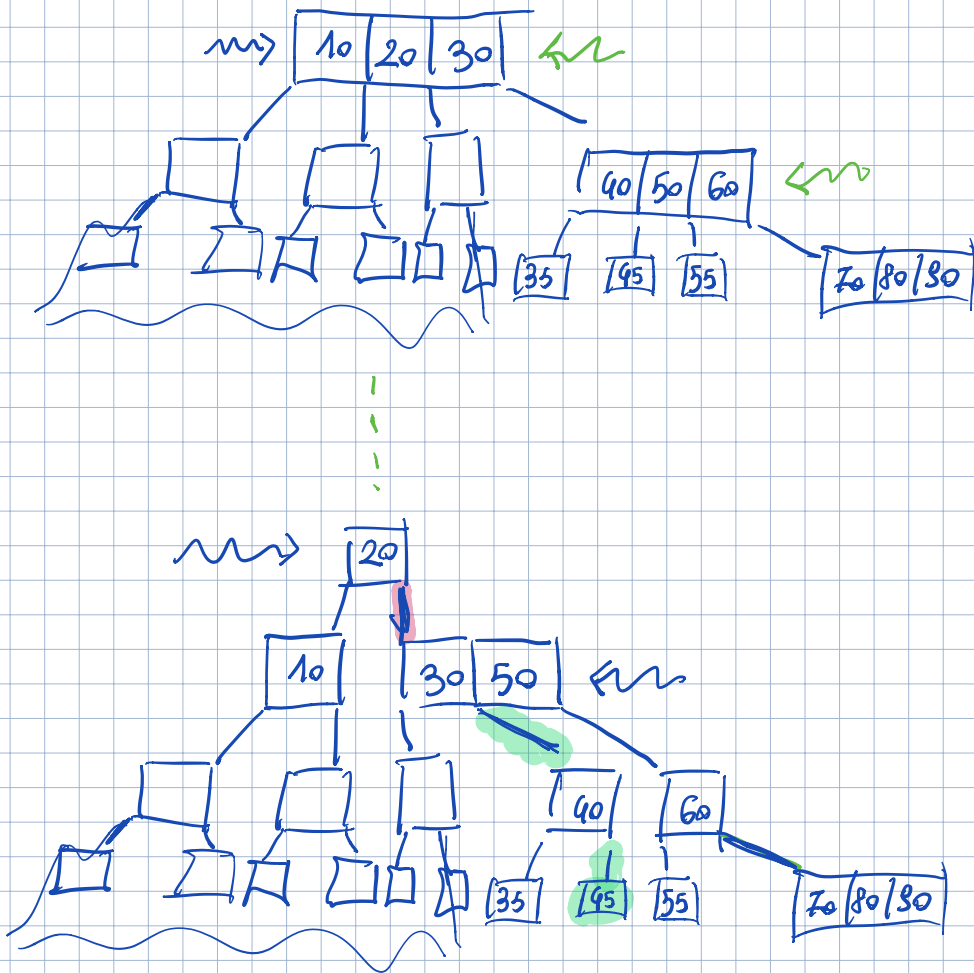
$h=3$



$2t-1$ sempre dispari

$$2t-1 = \underbrace{(t-1)} + \underbrace{1} + \underbrace{(t-1)}$$

$k = 48$



Create albero vuoto



$x.m$

$x.leaf = true$

BtreeCreate (T, t) {

$x \leftarrow \text{Allocate new node } (t)$

$x.leaf \leftarrow true$

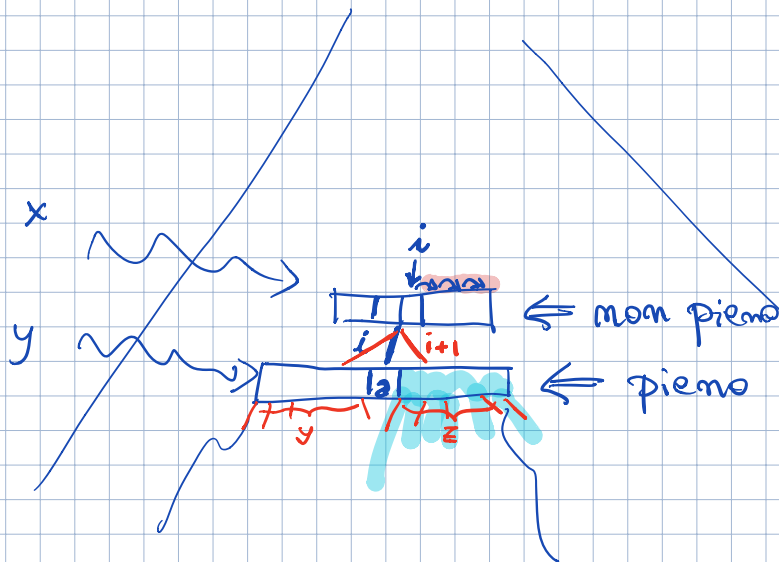
$x.m \leftarrow 0$

$T.root \leftarrow \text{DiskWrite}(x)$

return T

}

$x.key$ di $lung. 2t-1$
 $x.c$ di $lung. 2t$



x e y in RAM

y i -esimo figlio di x

$y.m = 2t-1$

Btree Split (x, y, i, t) {

 % x, y in RAM

 % y i -esimo figlio

 % y pieno e

 % x non pieno

$z \leftarrow \text{Allocate new node } (t)$

$z.\text{leaf} \leftarrow y.\text{leaf}$

$z.m \leftarrow t - 1$

for ($j \leftarrow 1$ to $t - 1$) {

$z.\text{key}_j \leftarrow y.\text{key}_{t+j}$

}

if (not $y.\text{leaf}$) {

 for ($j \leftarrow 1$ to t) {

$z.c_j \leftarrow y.c_{t+j}$

 }

}

$y.m \leftarrow t - 1$

for ($j \leftarrow x.m$ down to i) {

$x.\text{key}_{j+1} \leftarrow x.\text{key}_j$

}

$x.key_i \leftarrow y.key_t$

for ($j \leftarrow x.m + 1$ downto $i + 1$)

$x.c_{j+1} \leftarrow x.c_j$

}

$x.c_{i+1} \leftarrow z$

$x.m \leftarrow x.m + 1$

Disk Write (x)

Disk Write (y)

Disk Write (z)

}

Costo

CPU $O(t)$

R/W $\Theta(1)$ m.to di R/W